

Experiment No. 0

Name of Experiment : Introduction

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What is Python?

Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.

It is used for:

- web development (server-side),
- software development,
- mathematics,
- System scripting.

What can Python do?

- Python can be used on a server to create web applications.
- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

Why Python?

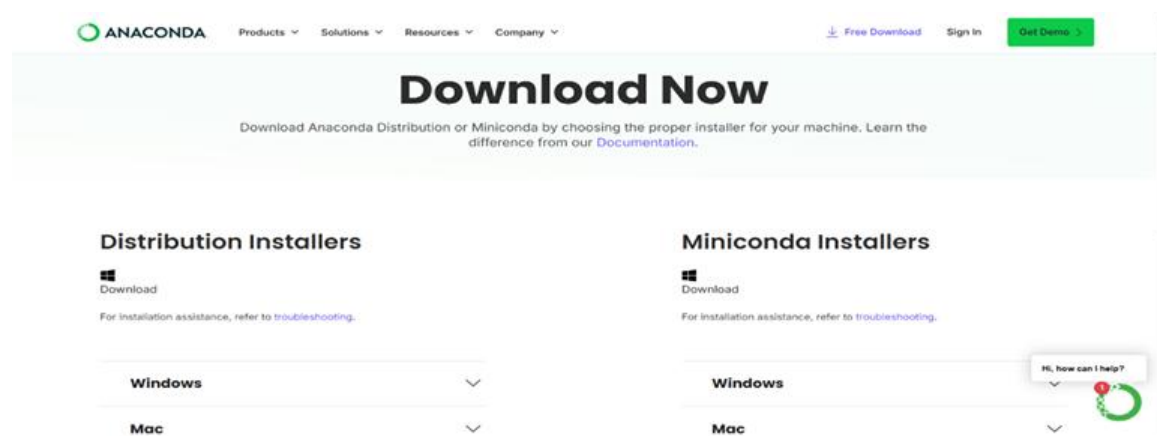
- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.

- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Python Installation Steps:

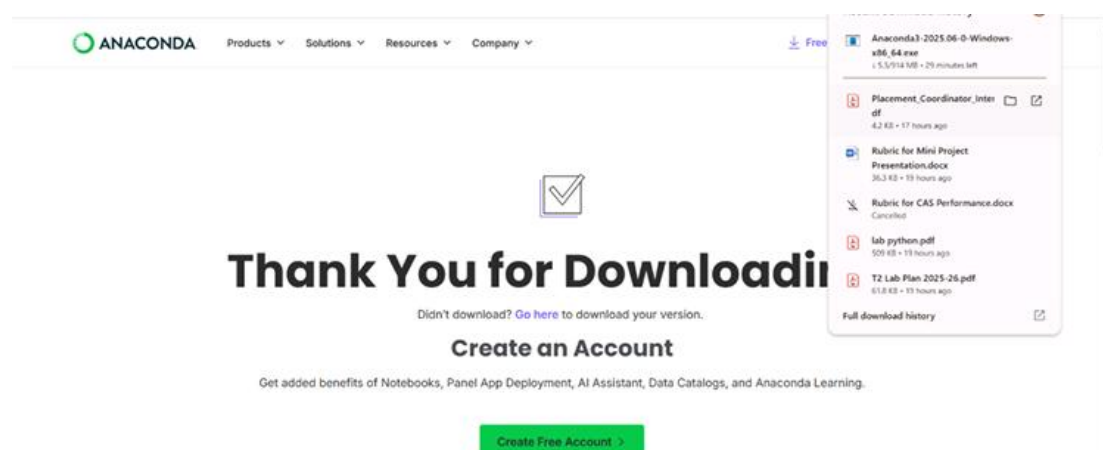
Step 1: Download the Anaconda Installer

Go to the official Anaconda website and navigate to the download page. Choose the installer that matches your operating system. For this setup, we are selecting the Windows version.



Step 2: Run the Installer

Once the file (e.g., Anaconda3-2025.06-0-Windows-x86_64.exe) finishes downloading, double-click it to start the installation.



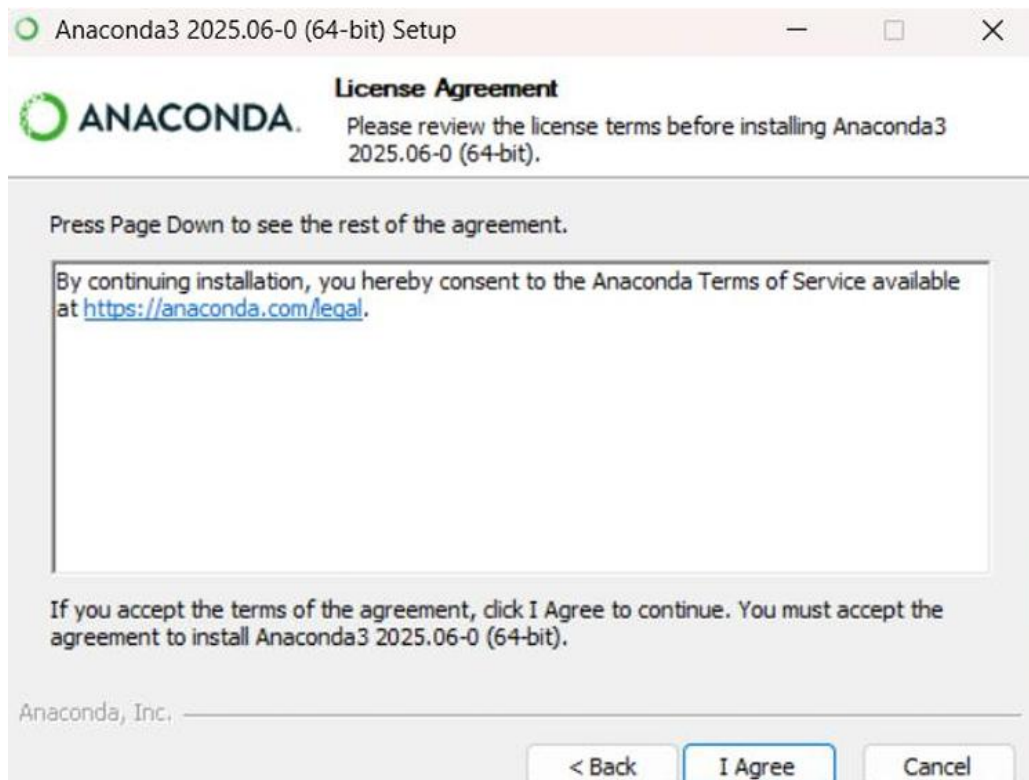
Step 3: Welcome Screen

The installer begins with a Welcome window. It is advised to close all other programs so that any system file updates happen smoothly. Click **Next** to proceed.



Step 4: Accept the License Agreement

You will be shown Anaconda's Terms of Service. Read and accept the license by clicking **I Agree** to continue the installation.

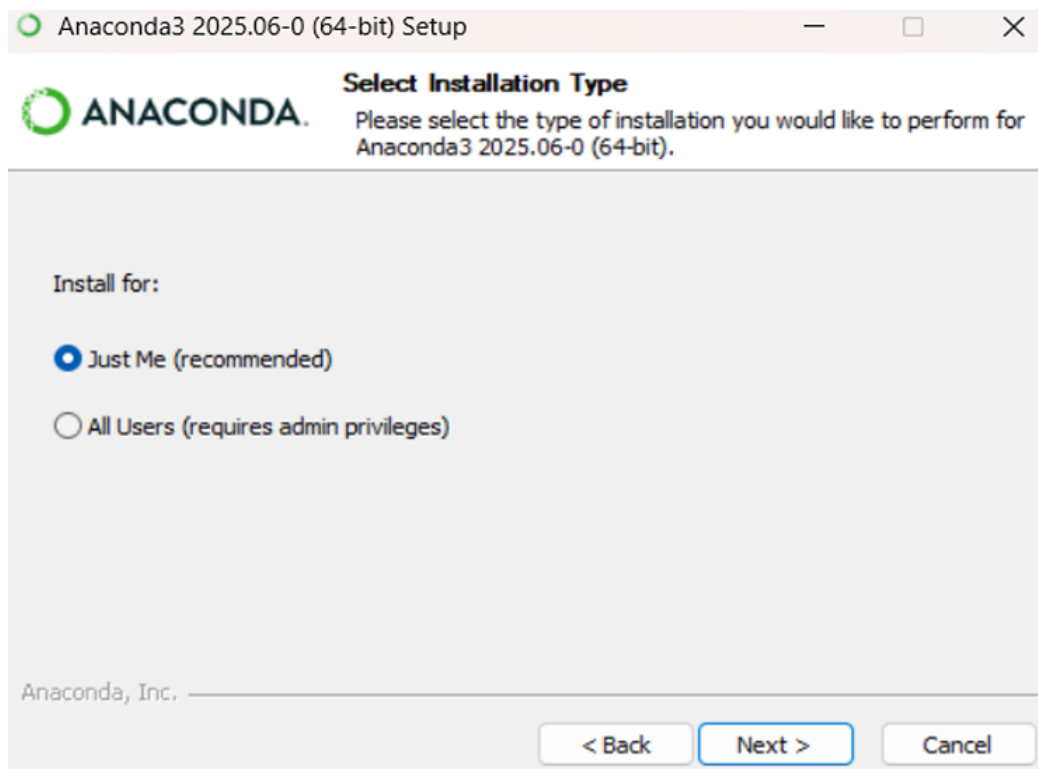


Step 5: Select Installation Type

Choose between **Just Me** and **All Users**:

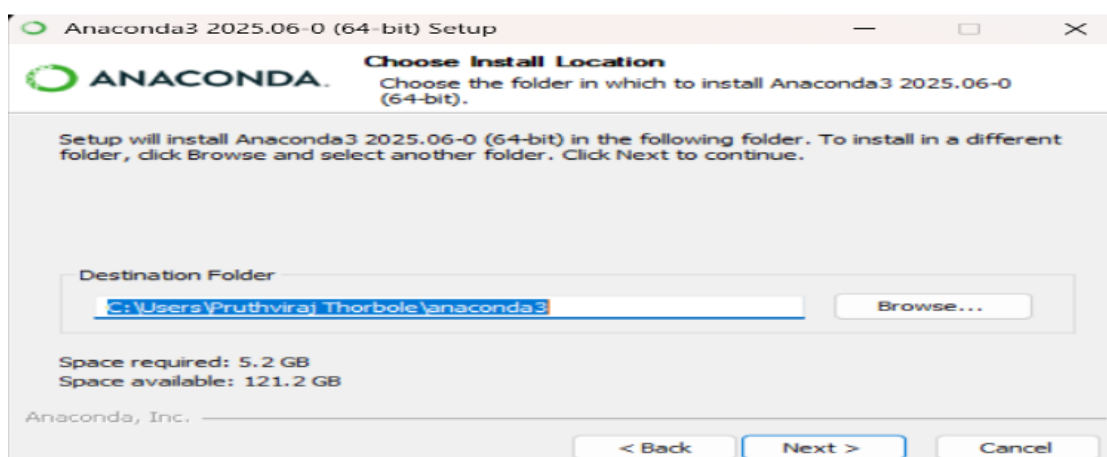
- **Just Me**: Installs Anaconda for the current user only (recommended).
- **All Users**: Installs Anaconda for all accounts on the system but requires admin access.

Select **Just Me** and click **Next**.



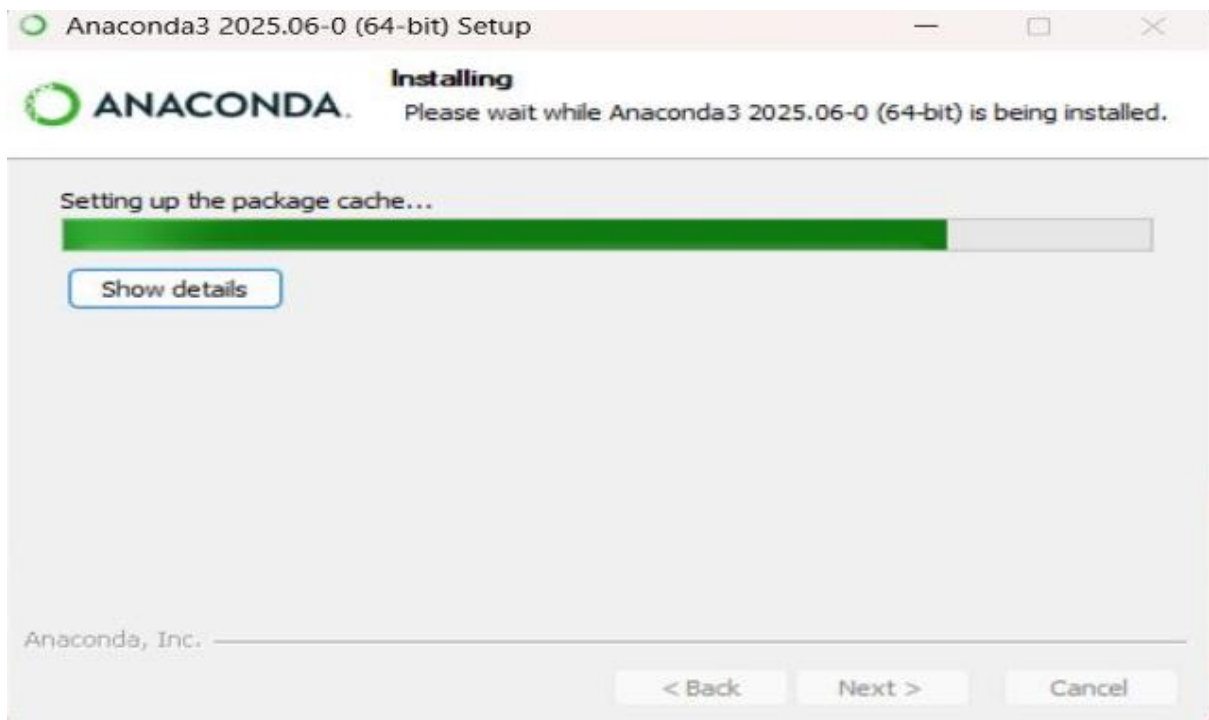
Step 6: Choose Install Location

Pick the folder where Anaconda will be installed. By default, it installs in your user folder (e.g., C:\Users\YourName\anaconda3). Confirm the location and click **Next**.



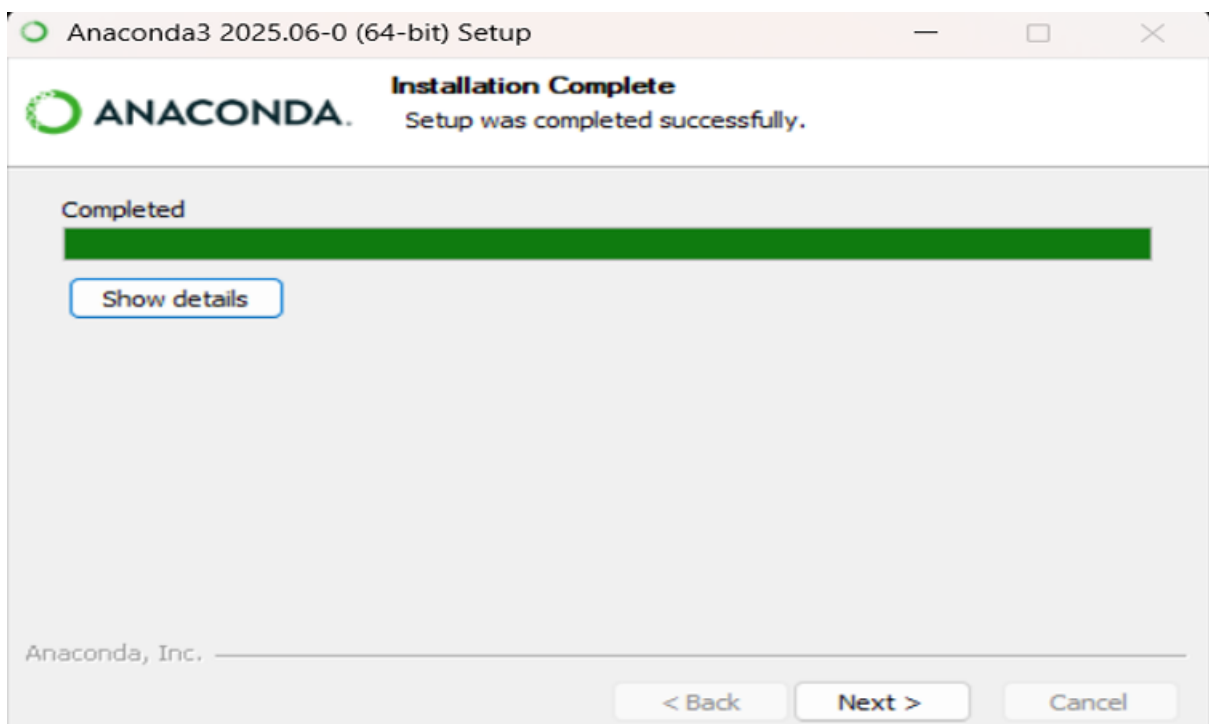
Step 7: Installation Process

The installer starts copying files and setting up all the packages. A progress bar will display the installation status.



Step 8: Installation Complete

Once all files are successfully installed, an “Installation Complete” message will appear. Click **Next** to move to the final steps.



Step 9: Anaconda Cloud Information

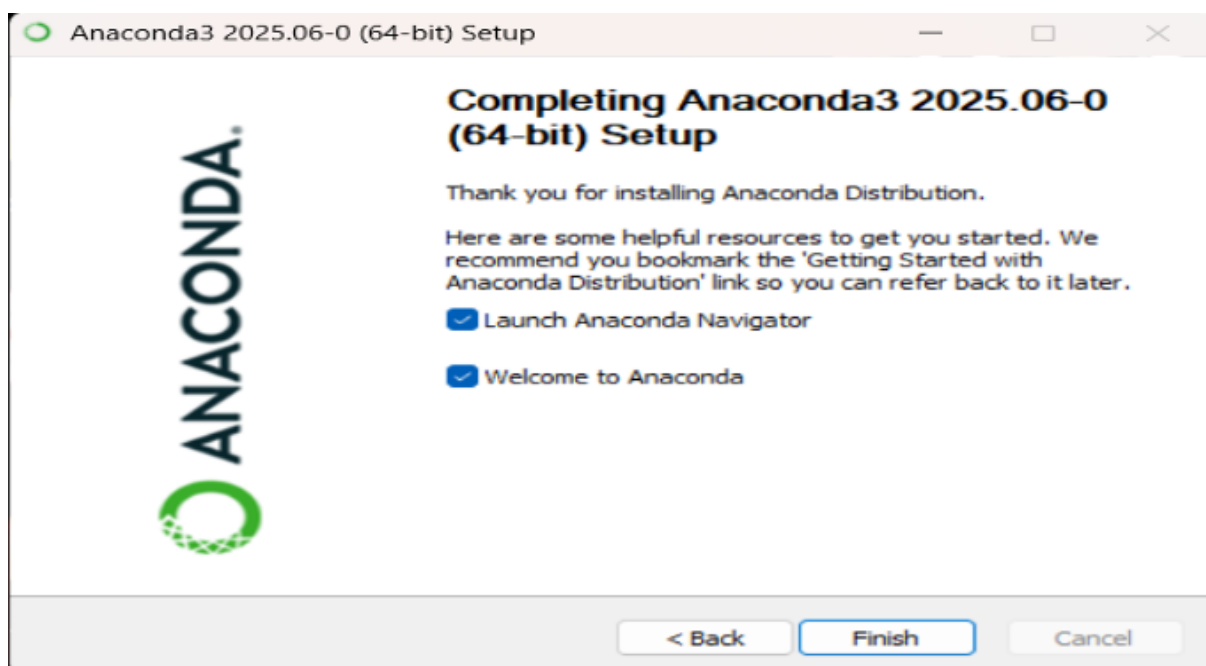
You will see a screen about Anaconda Cloud and online Jupyter notebooks. Click **Next** to continue.



Step 10: Final Setup

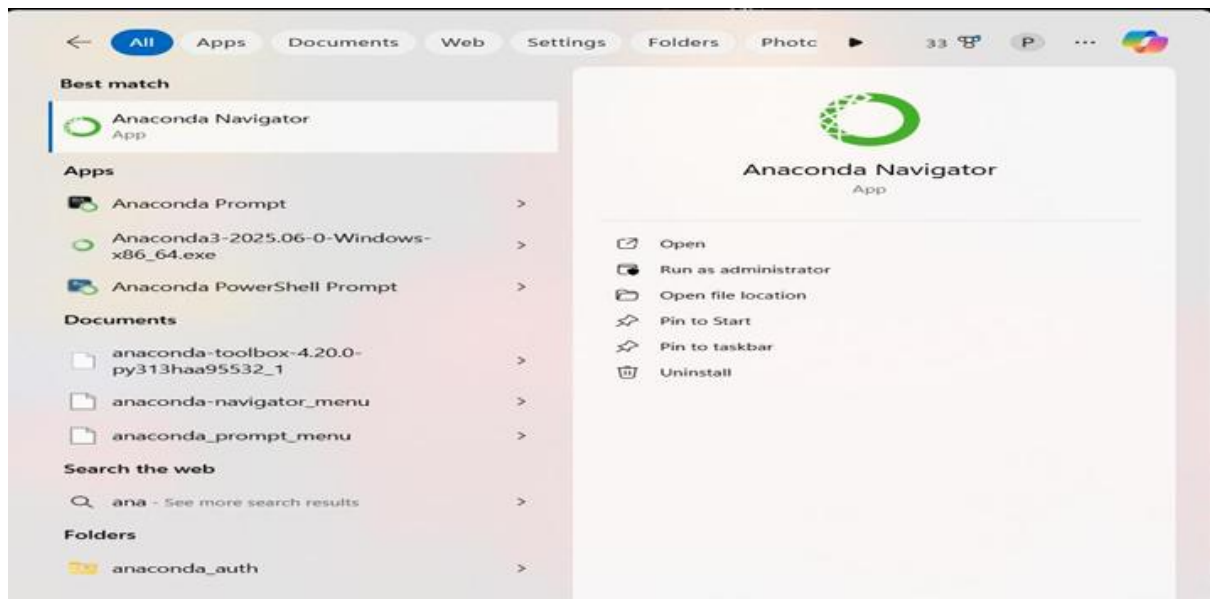
The final screen confirms that installation is finished. You can select:

- Launch Anaconda Navigator
 - View Welcome to Anaconda resources
- Click **Finish** to exit.



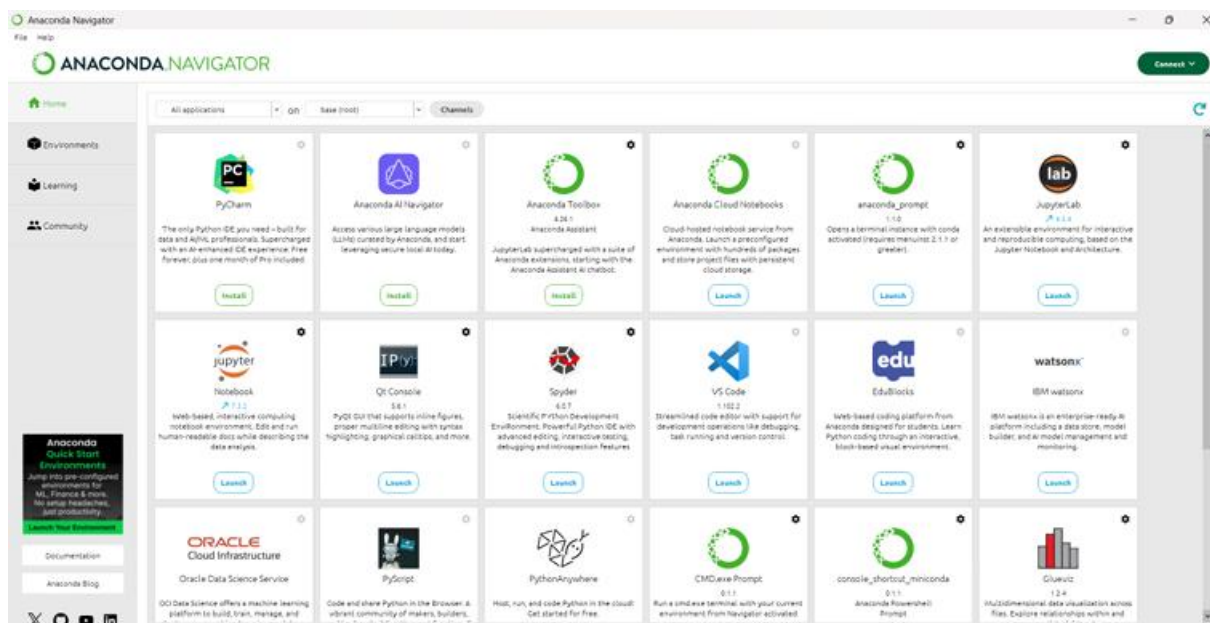
Step 11: Open Anaconda Navigator

Search for “Anaconda Navigator” in the Windows search bar and open it. This tool is your main control center for Anaconda environments and applications.



Step 12: Open Jupyter Notebook or JupyterLab

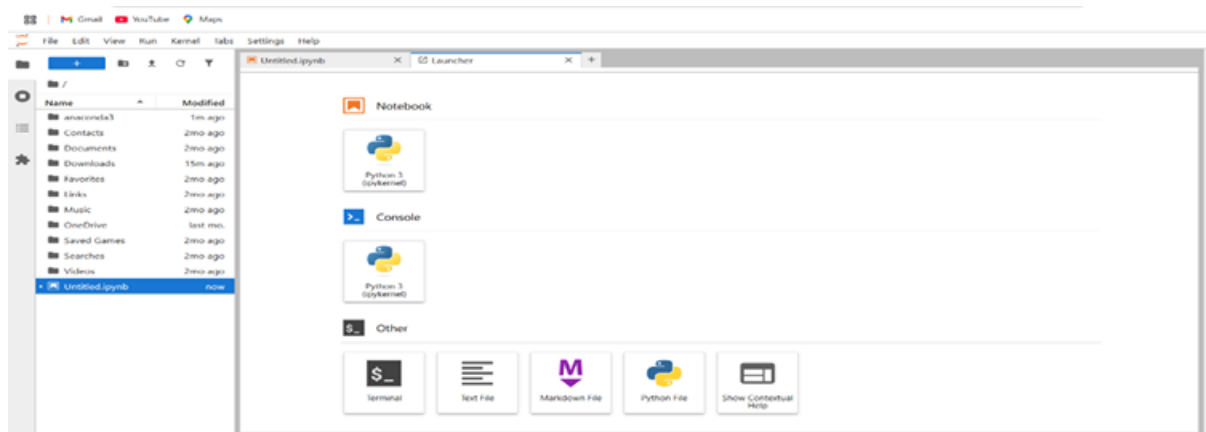
From Navigator’s home screen, find **Jupyter Notebook** or **JupyterLab** and click **Launch**.



Step 13: Create a New Notebook

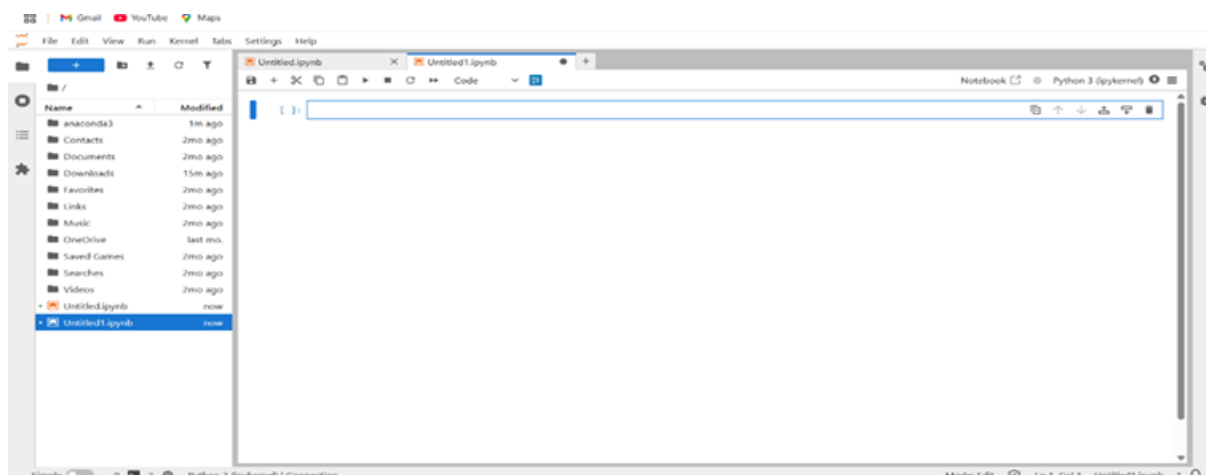
Your browser will open the Jupyter interface showing your files. To create a new notebook:

- In **JupyterLab**, use the “Launcher” tab.
- In classic **Jupyter Notebook**, use the “New” dropdown.
Choose **Python 3 (ipykernel)**.



Step 14: Start Writing Python Code

A new notebook opens (e.g., Untitled1.ipynb). Each notebook has cells where you can type Python code. Press **Shift + Enter** to run the code, and the output will appear right below the cell.



Conclusion

I have now installed Anaconda on Windows and can start working in a Python environment suitable for data science and machine learning. Tools like **Anaconda Navigator** and **Jupyter Notebook** will help me write, test, and execute your Python programs efficiently.

YouTube link by S D Khatawkar:

1. Anaconda Installation on Windows 10: <https://youtu.be/M7Q8v9OggDU>
2. Working with jupyter notebook : https://youtu.be/pro5x_02FfU